

Mirwais Morrad

🌐 [Portfolio](#) | 📧 [Mirwais-Morrad](#) | 💻 [mirwais001](#) | ✉ mirwaismorrad1@gmail.com | 📞 +1 (249) 875-0432

PROFESSIONAL SUMMARY

Computing graduate from Queen's University with an option in Security, bringing hands-on experience building real-world software systems and a strong foundation in cybersecurity, cryptography, networking, and secure application development. Proficient in Python, Java, C, and Git, with practical experience across full-stack development, secure protocol design, and software quality assurance. Actively pursuing CompTIA Security+ and Network+ certifications, a fast learner who picks up new systems quickly and is always ready to be trained and contribute from day one.

EDUCATION & ACHIEVEMENTS

Queen's University

Kingston, ON

Bachelor of Computing (Honours), Security Option | GPA: 3.51/4.0

2022 – 2026

- **Relevant Courses:** Intro to Cybersecurity, Cryptography, Software Quality Assurance, Software Architecture, Computer Networks, Intro to Artificial Intelligence, Operating Systems, Algorithms and Data Structures

SKILLS

Technical Skills: Python, Java, C, Haskell, Prolog, HTML/CSS, SQL, SQLite, Flask, Git, Figma, Wireshark, Cryptography, Network Security, Operating Systems (Windows, Linux), Microsoft Office (Word, PowerPoint, Excel, Teams, Outlook)

Soft Skills: Team Leadership, Cross-Cultural Communication, Staff Training and Onboarding, Customer Service, Adaptability, Multilingual (English, Mandarin, Farsi/Dari, Pashto, Urdu, Hindi), Technical Writing, Time Management

TECHNICAL PROJECTS

SecureShare — CISC 468: Cryptography [\[visit\]](#)

Winter 2026

- In a group of 2, built a peer-to-peer encrypted file sharing CLI application with two fully interoperable clients in Python and Java. Features include mDNS-based peer discovery, mutual RSA-2048 authentication, AES-256-GCM encrypted transport with perfect forward secrecy via ephemeral Diffie-Hellman, a password-protected local vault (PBKDF2 + AES-GCM), signed manifest-based offline file redistribution, and automated key rotation with KEY_MIGRATION broadcasting. Tested with 150 unit and interoperability tests across both clients.
- **Skills developed:** applied cryptography (RSA-PSS, AES-GCM, HKDF, PBKDF2, DH key exchange), dual-language system design (Python and Java interoperability), secure protocol design, multi-threaded networking, and test-driven development.

Password Strength Analyzer — Personal Cybersecurity Project [\[visit\]](#)

Winter 2026

- Built and deployed a Flask web application that evaluates password strength using entropy scoring (zxcvbn), 6 rule-based pattern checks, a Have I Been Pwned k-anonymity breach lookup, and a RockYou wordlist check loaded in memory. Secured with rate limiting, HTTP security headers, and a no-logging policy.
- **Skills developed:** secure API design, entropy-based analysis, privacy-preserving breach detection (k-anonymity), Flask app factory pattern, rate limiting with Flask-Limiter, and frontend JavaScript with async/await.

Phishing URL Detector — Personal Cybersecurity Project [\[visit\]](#)

Winter 2026

- Built and deployed a Flask web application that analyzes URLs for phishing indicators using a 17-rule heuristic scoring engine and three external threat intelligence feeds (PhishTank, URLhaus, Scamwave). Returns an explainable risk score (Low/Medium/High) without ever fetching the submitted URL, eliminating SSRF risk entirely.
- **Skills developed:** heuristic scoring system design, threat intelligence integration with caching and TTL management, SSRF prevention, pytest test suite covering all 17 detection rules, and securing a JSON API with input validation and security headers.

Other Projects: Rental Management Platform [\[visit\]](#) | GNUstep Architectural Analysis [\[visit\]](#)

WORK EXPERIENCE

Front Desk Representative

May 2023 – Present

Queen's University, Residence Life and Services | Kingston, ON

- Greeted and assisted hundreds of students, residents, and guests daily across four front desks, handling inquiries, resolving concerns, managing appointment bookings, writing shift logs, and handling 50+ emails daily, ensuring every person felt acknowledged and supported promptly.

Sales Associate, Seasonal Department

Feb 2023 – May 2025

The Home Depot | Kingston, ON

- Worked as a part-time Sales Associate at The Home Depot, helping customers find their needed products.